

The Telegraph Model of Gene Expression: A Triangular Cross-Validation of Stochastic Simulation

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Abstract

Gene expression is a stochastic process: even genetically identical cells in the same environment produce very different numbers of messenger RNA (mRNA) molecules. The telegraph model explains the main source of this variability. It treats a gene as a two-state (ON/OFF) switch that transcribes mRNA only while active. We study the telegraph model through three algebraically equivalent descriptions, all following from the same chemical master equation, and check that their independent implementations agree: we (i) generate stochastic trajectories with the Gillespie stochastic simulation algorithm (SSA); (ii) derive and solve a closed system of ordinary differential equations (ODEs) for the first and second moments; and (iii) draw independent samples directly from the analytic steady-state distribution, using the classic Beta–Poisson law together with an explicit Bernoulli gene state. The main contribution is the implementation cross-validation that links the three. The code is available at <https://github.com/zakhar-guzii/telegraph-model-gene-expression>.

1 Introduction

Gene expression is variable. Two cells with the same DNA, grown together in the same environment, can have different numbers of copies of the same mRNA or protein [6, 7]. This difference between cells is called phenotypic noise. It happens naturally because of random molecular interactions. Experimentally this variability separates into two parts [6]: *extrinsic* noise, caused by differences between cells—their size, cell-cycle stage, or the pool of available polymerases—and *intrinsic* noise, the randomness of the transcription and degradation events themselves, which persists even between two identical copies of a gene inside the same cell.

Inside a cell the molecules involved—a single copy of the gene, a few transcription factors, and a few dozen mRNA molecules—are present in such small numbers that the discreteness and randomness of individual chemical events cannot be smoothed out. The telegraph model studied here is the simplest framework that describes this intrinsic component. It treats the promoter as a switch that moves at random between an active (ON) and an inactive (OFF) state [8]. While ON, it produces mRNA molecules through transcription in random bursts; each molecule is then degraded at a random time. As a result, the mRNA count is a discrete-valued stochastic process.

The telegraph model reproduces the bursty, *over-dispersed* mRNA distributions seen in single-molecule experiments, which count individual mRNAs inside single cells [5], and it can be analysed exactly [4]. A convenient way to measure this spread is the *Fano factor* $F = \sigma^2/\mu$ [9], the ratio

of the variance to the mean of the mRNA count. Pure Poisson production and decay, in which every event is independent, has variance equal to mean, giving the *Poisson baseline* $F = 1$. “Over-dispersed” means $F > 1$: the random switching of the gene adds extra variability on top of that baseline.

We treat the telegraph model as a case study in how a stochastic process can be described from several mathematically distinct viewpoints. We implement three approaches that look unrelated:

1. a stochastic simulation (the SSA), which follows a single cell through time;
2. a system of moment ODEs, which gives the population mean and spread deterministically;
3. a steady-state sampler, which describes the population after a long time.

The three are algebraically equivalent—all follow from the same chemical master equation—but their implementations share no code and fail in different ways. Agreement between them is therefore evidence against implementation defects, not evidence about the master equation they all inherit.

2 Model Formulation

The state of the system at time t is the pair

$$(G(t), R(t)), \quad G(t) \in \{0, 1\}, \quad R(t) \in \mathbb{N}, \quad (1)$$

where the binary variable G records whether the gene is OFF ($G = 0$) or ON ($G = 1$), and the integer R counts the number of mRNA molecules present. Four basic reactions define the dynamics of the system. Each reaction has a *propensity* a_j , the rate at which it fires at any given moment:

Reaction	Description	Effect	Propensity
R_1	Gene activation	$G : 0 \rightarrow 1$	$a_1 = k_{\text{on}} (1 - G)$
R_2	Gene inactivation	$G : 1 \rightarrow 0$	$a_2 = k_{\text{off}} G$
R_3	RNA synthesis	$R \rightarrow R + 1$	$a_3 = k_{\text{syn}} G$
R_4	RNA degradation	$R \rightarrow R - 1$	$a_4 = k_{\text{deg}} R$

All four rate constants are positive. The switching rates $k_{\text{on}}, k_{\text{off}}$ set how quickly the promoter turns on and off, and their ratio fixes the long-run fraction of time the gene is active; the factor G in a_3 shuts transcription off entirely when the gene is OFF; and k_{deg} is a per-molecule decay rate, so the total degradation propensity $k_{\text{deg}} R$ grows with the amount of mRNA present. Together the rates $(k_{\text{on}}, k_{\text{off}}, k_{\text{syn}}, k_{\text{deg}})$ and an initial condition (g_0, r_0) at time t_0 fully determine the law of the process.

Burst size. Once the gene is ON, the only reaction that can switch it off fires at the constant rate k_{off} , so the duration of an active period is exponential with mean $1/k_{\text{off}}$; during that period transcription fires at rate k_{syn} . The mean number of transcripts made per active period—the *burst size* B —is therefore

$$B = k_{\text{syn}} \mathbb{E}[\text{ON duration}] = \frac{k_{\text{syn}}}{k_{\text{off}}}. \quad (2)$$

Bursts of this kind are observed directly in single cells [10, 5], and the exact consequences of promoter switching for the mRNA distribution, including the limits recovered in Section 4.2, are derived in [11].

Because each waiting time is exponential and depends only on the current state, $(G(t), R(t))$ is a continuous-time Markov chain (CTMC). Its distribution $P_t(g, r) = \Pr\{G(t) = g, R(t) = r\}$ evolves according to the chemical master equation [14]

$$\begin{aligned} \frac{d}{dt} P_t(g, r) = & k_{\text{on}} [\delta_{g,1} P_t(0, r) - \delta_{g,0} P_t(g, r)] \\ & + k_{\text{off}} [\delta_{g,0} P_t(1, r) - \delta_{g,1} P_t(g, r)] \\ & + k_{\text{syn}} \delta_{g,1} [P_t(1, r-1) - P_t(1, r)] \\ & + k_{\text{deg}} [(r+1) P_t(g, r+1) - r P_t(g, r)], \end{aligned} \quad (3)$$

with the convention $P_t(g, -1) = 0$. Equation (3) is the basis for all three approaches below: the SSA samples trajectories whose distribution solves it, the moment ODEs are derived from it exactly, and the steady-state sampler draws from its $t \rightarrow \infty$ fixed point.

3 Methods

3.1 Stochastic Simulation (Gillespie SSA)

The *stochastic simulation algorithm* (SSA) of Gillespie [1, 2, 3] generates statistically exact sample paths of a CTMC, making no approximation to the master equation (3). Its key idea is that, given the current state, two questions separate: when does the next reaction occur, and which reaction is it? With $a_0 = \sum_{j=1}^4 a_j$ the *total propensity*, the waiting time is $\tau \sim \text{Exp}(a_0)$ and reaction j fires with probability a_j/a_0 .

Our implementation draws these by inversion, using two independent uniforms $r_1, r_2 \sim \text{U}(0, 1)$ per step from a seeded `numpy.random` generator [17], one seed per experiment:

$$\tau = \frac{1}{a_0} \ln \frac{1}{r_1}, \quad j = \min \left\{ k : \sum_{i=1}^k a_i > r_2 a_0 \right\}. \quad (4)$$

The time is then advanced by τ , the state updated by reaction j , the reaction propensities recomputed, and the loop repeats for a fixed number of events n_{sim} . If the total propensity reaches $a_0 = 0$, no reaction can fire, so the state is absorbing and is held fixed afterwards.

A single trajectory is the history of one cell. To obtain population statistics, we run N_{rep} independent trajectories with identical parameters and average over them. Each trajectory advances by its own random time increments, so the same event index occurs at different times in different trajectories. We therefore resample every trajectory onto a shared uniform time grid by a zero-order hold before averaging; the state is piecewise constant between events, so this introduces no error. The empirical moments are then ensemble averages at each grid time, for example

$$\widehat{\mu}_R(t) = \frac{1}{N_{\text{rep}}} \sum_{i=1}^{N_{\text{rep}}} R_i(t), \quad \widehat{\mu}_G(t) = \frac{1}{N_{\text{rep}}} \sum_{i=1}^{N_{\text{rep}}} G_i(t), \quad (5)$$

with sample variances and the sample gene–RNA covariance defined analogously.

Simulation horizon. The loop counter n_{sim} bounds the number of *events*, not the elapsed time, so an ensemble has no common end time. We take the shared grid no further than the earliest-finishing trajectory, $t_{\text{end}} = \min_i t_i^{\text{last}}$; extending past that point would freeze the exhausted trajectories at their final state and bias the ensemble toward the states those cells held when their event budget

ran out. Each figure ends at its own seed-dependent time, in every case a large multiple of the relaxation time.

A typical sample path is shown in Figure 1. The mRNA count climbs in a burst while the gene is ON and decays roughly exponentially once it switches OFF, which is the mechanism behind over-dispersion; and the decay is slow enough that the count is still well above zero when the next ON period begins, so a cell can be found with $G = 0$ and $R > 0$. The steady-state sampler of Section 3.3 reproduces this property.

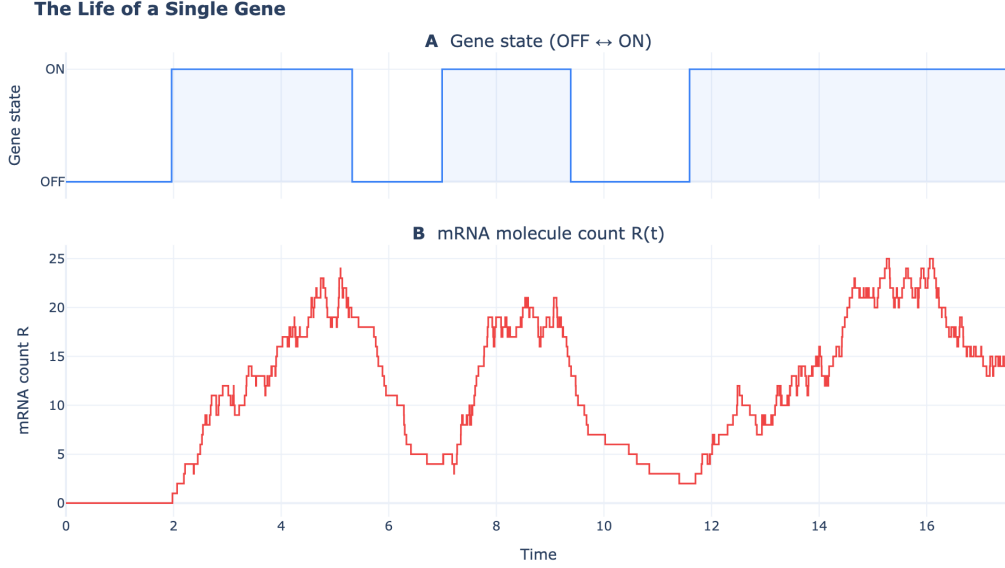


Figure 1: A single SSA sample path at the reference parameters of Section 4. Top: the gene state switches between OFF and ON at random times. Bottom: the mRNA count rises in bursts during ON periods and decays between them. Averaging many such trajectories according to (5) gives the ensemble moments compared with the moment ODEs in Section 4.3.

3.2 Moment ODEs

The SSA describes individual cells. The population-level statistics—the *moments*—obey deterministic equations instead. A moment is an expectation: the mean $\mu_R = \mathbb{E}[R]$ is the average mRNA level, the variance $\sigma_R^2 = \mathbb{E}[R^2] - \mathbb{E}[R]^2$ measures the noise, and the covariance $C_{RG} = \mathbb{E}[RG] - \mathbb{E}[R]\mathbb{E}[G]$ measures how gene activity and mRNA level vary together.

Applying the standard moment rule for a chemical master equation—the expected rate of change of any observable is the sum over reactions of each propensity times the change it makes—to $f = G, R, R^2, GR$ in turn yields four coupled equations for the telegraph model [16]:

$$\frac{d\mu_G}{dt} = k_{\text{on}} (1 - \mu_G) - k_{\text{off}} \mu_G, \quad (6)$$

$$\frac{d\mu_R}{dt} = k_{\text{syn}} \mu_G - k_{\text{deg}} \mu_R, \quad (7)$$

$$\frac{d\sigma_R^2}{dt} = 2k_{\text{syn}} C_{RG} - 2k_{\text{deg}} \sigma_R^2 + k_{\text{syn}} \mu_G + k_{\text{deg}} \mu_R, \quad (8)$$

$$\frac{dC_{RG}}{dt} = k_{\text{syn}} \sigma_G^2 - (k_{\text{on}} + k_{\text{off}} + k_{\text{deg}}) C_{RG}. \quad (9)$$

Exact closure. In a general nonlinear reaction system the equation for each moment depends on higher moments, so the system never closes and one has to add approximate closure schemes [12, 15]. Here it closes *exactly*. The only possible nonlinearity is the variance of the gene state, but because G is binary,

$$\sigma_G^2 = \mathbb{E}[G^2] - \mu_G^2 = \mathbb{E}[G] - \mu_G^2 = \mu_G (1 - \mu_G), \quad (10)$$

which depends on the mean μ_G alone. Putting (10) into (9) turns the four equations (6)–(9) into a linear system in $(\mu_G, \mu_R, \sigma_R^2, C_{RG})$ whose solutions are the true moments of the master equation (3).

Implementation. The four right-hand sides (6)–(9), with σ_G^2 substituted by (10), are coded directly and passed to the explicit Runge–Kutta solver `scipy.integrate.solve_ivp` [18] (method RK45, the embedded Dormand–Prince pair [19]) from the initial conditions $\mu_G(t_0) = g_0$, $\mu_R(t_0) = r_0$, $\sigma_R^2(t_0) = C_{RG}(t_0) = 0$. Section 4.3 overlays the result on the SSA ensemble at the same rates and the same initial condition.

Steady-state Fano factor. The Fano factor at stationarity follows from the same four equations in three steps.

First, set the right-hand side of (9) to zero and use the closure (10) for σ_G^2 :

$$C_{RG}^{ss} = \frac{k_{\text{syn}} \sigma_G^2}{k_{\text{on}} + k_{\text{off}} + k_{\text{deg}}} = \frac{k_{\text{syn}} \mu_G^{ss} (1 - \mu_G^{ss})}{k_{\text{on}} + k_{\text{off}} + k_{\text{deg}}}. \quad (11)$$

Second, set the right-hand side of (8) to zero and solve for the variance:

$$2k_{\text{deg}} \sigma_R^2 = 2k_{\text{syn}} C_{RG}^{ss} + k_{\text{syn}} \mu_G^{ss} + k_{\text{deg}} \mu_R^{ss}. \quad (12)$$

Third, note that $k_{\text{syn}} \mu_G^{ss} = k_{\text{deg}} \mu_R^{ss}$ by (7), so the last two terms of (12) are equal. Substituting (11) and dividing through by μ_R^{ss} gives the Fano factor in closed form:

$$F^{ss} = \frac{\sigma_R^2}{\mu_R} \Big|_{ss} = 1 + \frac{k_{\text{syn}} k_{\text{off}}}{(k_{\text{on}} + k_{\text{off}})(k_{\text{on}} + k_{\text{off}} + k_{\text{deg}})}. \quad (13)$$

The additive 1 is the Poisson baseline. The second term is the excess contributed by promoter switching. It is strictly positive whenever $k_{\text{off}} > 0$, decays like $1/(k_{\text{on}} + k_{\text{off}})$ at fixed ratio, and vanishes in two limits: as $k_{\text{off}} \rightarrow 0$, where the gene is always ON and the mRNA count becomes a pure birth–death process, and as $k_{\text{on}}, k_{\text{off}} \rightarrow \infty$, where a promoter that switches very fast averages out.

Setting the time derivatives in (6)–(7) to zero in the same way gives the exact stationary means used as the reference throughout Section 4:

$$\mu_G^{ss} = \frac{k_{\text{on}}}{k_{\text{on}} + k_{\text{off}}}, \quad \mu_R^{ss} = \frac{k_{\text{syn}}}{k_{\text{deg}}} \mu_G^{ss}. \quad (14)$$

3.3 Steady-State Sampler

As $t \rightarrow \infty$ the telegraph model forgets its initial condition and converges to a unique stationary distribution $P_\infty(g, r)$. Instead of simulating a long transient to reach it, we can sample from P_∞ directly. Peccoud and Ycart [4] showed that at steady state the mRNA count follows the classic *Beta–Poisson* law: it is Poisson, but with a mean that is itself random from cell to cell. Let the *mixing fraction* $p \in [0, 1]$ be the long-run share of time a cell’s gene spends ON. Across a population p

is Beta-distributed, and given p the mRNA count is Poisson with mean $\lambda = cp$, where $c = k_{\text{syn}}/k_{\text{deg}}$ is the *transcription efficiency coefficient*: the mean mRNA level a permanently active gene would reach. With the switching rates $a = k_{\text{on}}/k_{\text{deg}}$ and $b = k_{\text{off}}/k_{\text{deg}}$ (so k_{deg} fixes the unit of time), the stationary mRNA law is therefore

$$R \sim \text{Poisson}(cp), \quad p \sim \text{Beta}(a, b). \quad (15)$$

Averaging the Poisson kernel over the Beta density gives $P_{\infty}(r)$ in closed form, as a confluent hypergeometric expression [4]. Because the mixture (15) is that distribution, sampling the mixture samples P_{∞} exactly, using three draws per sample and no discretisation of the state space.

Our sampler draws the gene state as an explicit Bernoulli(p) so we can also estimate the joint moments of (G, R) . For each of the N_{rep} samples it draws

$$p \sim \text{Beta}(a, b), \quad f(p) = \frac{\Gamma(a+b)}{\Gamma(a)\Gamma(b)} p^{a-1}(1-p)^{b-1}, \quad (16)$$

$$G \mid p \sim \text{Bernoulli}(p), \quad \Pr(G = 1 \mid p) = p, \quad \Pr(G = 0 \mid p) = 1 - p, \quad (17)$$

$$R \mid p \sim \text{Poisson}(\lambda), \quad \lambda = \frac{k_{\text{syn}}}{k_{\text{deg}}} p. \quad (18)$$

The mRNA count R is conditioned on the mixing fraction p , *not* on the current gene state G . Given p , the gene state and the mRNA count are drawn independently, so a cell can have $G = 0$ and still $R > 0$ —as in a real cell, where mRNA made during the last active period survives past the switch to OFF.

The joint construction. Peccoud and Ycart [4] derived the *marginal* law of R , whereas the checks below need $\widehat{\sigma}_G^{ss}$ and $\widehat{C}_{RG}^{ss}$, which are properties of the joint law. Coupling G to the same mixing fraction p that drives the Poisson mean supplies that joint law, and it reproduces the covariance C_{RG}^{ss} of (11), obtained independently in Section 3.2 by setting (9) to zero.

Because G and R are conditionally independent given p ,

$$\mathbb{E}[RG] = \mathbb{E}[\mathbb{E}[R \mid p] \mathbb{E}[G \mid p]] = c \mathbb{E}[p^2], \quad \mathbb{E}[R] \mathbb{E}[G] = c \mathbb{E}[p]^2, \quad (19)$$

so that $C_{RG} = c \text{Var}(p)$. Substituting the Beta variance and rewriting each factor in terms of the original rates then gives

$$\begin{aligned} C_{RG} &= c \cdot \frac{ab}{(a+b)^2(a+b+1)} = \frac{k_{\text{syn}}}{k_{\text{deg}}} \cdot \frac{\mu_G^{ss}(1-\mu_G^{ss})}{(k_{\text{on}} + k_{\text{off}} + k_{\text{deg}})/k_{\text{deg}}} \\ &= \frac{k_{\text{syn}} \mu_G^{ss}(1-\mu_G^{ss})}{k_{\text{on}} + k_{\text{off}} + k_{\text{deg}}}, \end{aligned} \quad (20)$$

using $ab/(a+b)^2 = \mu_G^{ss}(1-\mu_G^{ss})$, $a+b+1 = (k_{\text{on}} + k_{\text{off}} + k_{\text{deg}})/k_{\text{deg}}$ and $c = k_{\text{syn}}/k_{\text{deg}}$. This matches (11) term for term, so the construction reproduces the joint moments of the master equation. The match is an algebraic identity rather than an empirical result: the numerical check in Section 4.4 tests the Monte Carlo estimator, not the construction itself.

Boundary behaviour. As the switching rates become slow compared with degradation ($a, b \rightarrow 0$), the Beta(a, b) density concentrates its mass at the endpoints $p = 0$ and $p = 1$, and the mixture separates into a single Bernoulli trial for the gene state and, when ON, an ordinary Poisson count for the mRNA. The limiting cases $k_{\text{off}} = 0$ and $k_{\text{deg}} = 0$ are treated in Section 4.4.

The joint stationary law is the average of the conditional distributions over the mixing fraction, $P_\infty(g, r) = \int_0^1 \Pr(G = g \mid p) \Pr(R = r \mid p) f(p) dp$; drawing N_{rep} triples (p, G, R) and averaging gives unbiased Monte Carlo estimates of the steady-state moments $\widehat{\mu}_G^{ss}, \widehat{\mu}_R^{ss}$, and so on.

4 Experiments and Results

Unless stated otherwise, the experiments use the symmetric *reference parameters* $k_{\text{on}} = 0.5$, $k_{\text{off}} = 0.5$, $k_{\text{syn}} = 20.0$, $k_{\text{deg}} = 1.0$, with $N_{\text{rep}} = 1000$ trajectories and initial condition $(g_0, r_0) = (0, 0)$, giving $\mu_G^{ss} = 0.5$ and $\mu_R^{ss} = 10$ by (14). The parameter-exploration runs of Section 4.2 are the one exception: they vary rates around a slower base of $k_{\text{on}} = k_{\text{off}} = 0.1$, stated with each run in Table 1. All three parts of the validation are implemented in Python; see the repository linked above.

4.1 The Validation Triangle

The three descriptions of Section 3 must agree wherever their domains overlap. We state those overlaps as the three edges of a *validation triangle* and test each one numerically below.

Edge 1 (SSA $\leftrightarrow$ moment ODEs, finite time): the empirical averages (5) are sample means over i.i.d. trajectories, so they converge to the expectations $\mu_G(t), \mu_R(t), C_{RG}(t)$ solving (6)–(9), and at finite N_{rep} must follow the ODE curves up to an error of order $N_{\text{rep}}^{-1/2}$. *Edge 2* (moment ODEs $\leftrightarrow$ steady-state sampler, long time): the ODEs relax to the fixed point (14), which the sampler must estimate at the same $N_{\text{rep}}^{-1/2}$ rate. *Edge 3* (SSA $\leftrightarrow$ steady-state sampler): a trajectory run to stationarity samples the same P_∞ as the sampler. Since both endpoints of this edge produce samples, Edge 3 can be tested as equality of the full laws, $\hat{P}_\infty^{\text{SSA}} = \hat{P}_\infty^{\text{smp}}$, rather than of their first two moments alone; Section 4.4 does so in total-variation distance.

4.2 Parameter Exploration

Before testing the three edges, we verify that the SSA accurately reproduces the closed-form Fano factor (13) across different parameter regimes. Table 1 isolates the effects of varying biological parameters and numerical settings.

As expected, switching speed modulates noise at a fixed mean (Figure 2): faster switching averages out the gene state, dropping σ_R^2 and C_{RG} and driving the Fano factor toward the Poisson baseline. Conversely, varying the burst size (2) alters the steady-state mean μ_R^{ss} . The expression runs demonstrate that a weaker promoter yields fewer molecules but a larger relative spread: the coefficient of variation σ_R/μ_R rises from 1.268 ± 0.012 in the high-expression run to 1.561 ± 0.006 in the low-expression run (Figure 3).

The steady state is robust to both initial conditions and numerical resolution. Across all nine runs in Table 1, the measured Fano factors agree with the analytical steady state (13) within 1.2 Monte Carlo standard errors (maximum relative deviation of 1.6%). The numerical experiments also confirm expected Monte Carlo error scaling: reducing the ensemble size N_{rep} from 1000 to 224 leaves $\hat{F}$ unbiased while widening its standard error predictably.

4.3 SSA vs. ODE: Moment Dynamics (Edge 1)

The direct test of Edge 1 is the overlay in Figure 4, where the SSA ensemble averages follow the deterministic ODE curves throughout the transient relaxation and into the steady state. The visible deviations are confined to the covariance curve—the noisiest of the three statistics—and shrink as

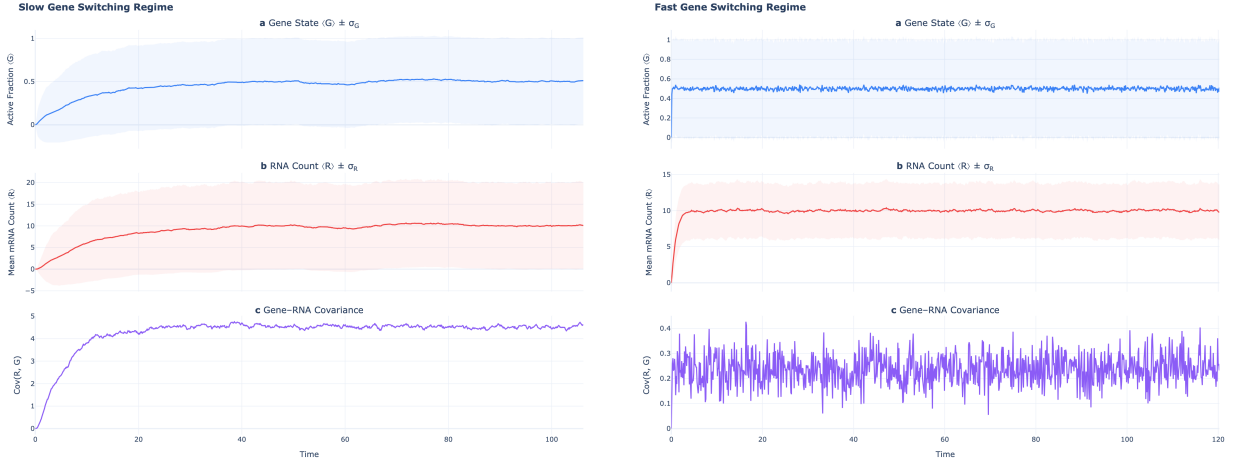


Figure 2: Switching speed reshapes noise at a fixed mean ($\mu_R \approx 10$). Slow switching (left, $k_{\text{on}} = k_{\text{off}} = 0.05$) produces long bursts, a wide $\pm\sigma_R$ band, and strong gene–RNA covariance. Fast switching (right, $k_{\text{on}} = k_{\text{off}} = 10$) averages the promoter out, leaving a narrow band and near-zero covariance.

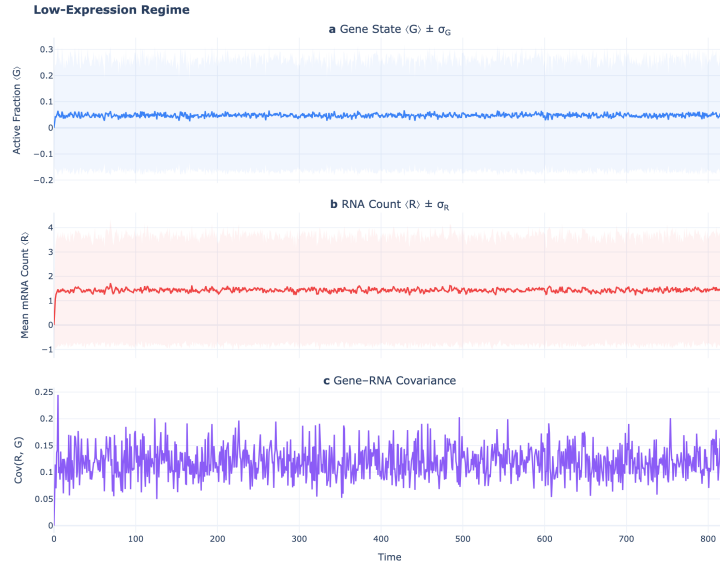


Figure 3: The low-expression regime ($k_{\text{on}} = 0.5$, $k_{\text{off}} = 10$, $k_{\text{syn}} = 30$), a weak promoter with burst size $B \approx 3$. The mean mRNA level is low and the relative spread correspondingly large; panel layout follows Figure 2.

Experiment	Parameters	F^{ss}	$\hat{F}$ (SSA)
<i>Noise at a fixed mean</i>			
Slow switching	$k_{\text{on}} = k_{\text{off}} = 0.05, k_{\text{syn}} = 20$	10.09	9.94 ± 0.24
Fast switching	$k_{\text{on}} = k_{\text{off}} = 10, k_{\text{syn}} = 20$	1.48	1.48 ± 0.02
<i>Burst size</i>			
High expression	$k_{\text{on}} = 0.5, k_{\text{off}} = 4, k_{\text{syn}} = 56$	10.05	9.90 ± 0.13
Low expression	$k_{\text{on}} = 0.5, k_{\text{off}} = 10, k_{\text{syn}} = 30$	3.48	3.49 ± 0.03
<i>Forgetting the initial condition</i>			
Base, from $(0, 0)$	$(g_0, r_0) = (0, 0)$	9.33	9.41 ± 0.16
From $(1, 15)$	$(g_0, r_0) = (1, 15)$	9.33	9.27 ± 0.08
<i>Numerical settings</i>			
Fewer events	$n_{\text{sim}} = 1000$	9.33	9.48 ± 0.21
Smaller ensemble	$N_{\text{rep}} = 224$	9.33	9.34 ± 0.46
Both	$N_{\text{rep}} = 224, n_{\text{sim}} = 5000$	9.33	9.40 ± 0.18

Table 1: The parameter-exploration runs. Unless stated otherwise, $k_{\text{deg}} = 1$, base switching rates are $k_{\text{on}} = k_{\text{off}} = 0.1$, $N_{\text{rep}} = 1000$, and $n_{\text{sim}} = 4000$. Each run maintains the reference steady-state mean except the expression runs. F^{ss} is the analytical Fano factor (13); $\hat{F} = \hat{\sigma}_R^2 / \hat{\mu}_R$ is the SSA estimate averaged over the final fifth of the simulation. The quoted $\pm$ represents the standard deviation across 10 independent seeds (the Monte Carlo error of a single run).

N_{rep} increases. The $\pm\sigma$ bands are wide because they represent true cell-to-cell variability rather than numerical error, perfectly matching the analytical spread predicted by the moment ODEs.

This agreement extends beyond the symmetric reference case: the SSA accurately tracks the deterministic dynamics under slow and fast switching, high and low expression, and various initial conditions. For instance, Figure 5 demonstrates that the exact tracking is preserved in the high-noise slow-switching regime.

While the overlays demonstrate qualitative agreement, Figure 6 rigorously quantifies the convergence rate, which Edge 1 predicts to be $\mathcal{O}(1/\sqrt{N_{\text{rep}}})$. We measure the root-mean-square (RMS) deviation across all grid points to ensure a robust, low-variance estimator. On log-log axes, the deviation is highly linear over three decades. A fit over this stable range yields a slope of -0.509 ± 0.010 , seamlessly matching the $-1/2$ scaling predicted by the Law of Large Numbers.

Edge 1 therefore holds across every regime tested. What converges is the estimate of the mean: the $\pm\sigma$ bands measure persistent cell-to-cell variability, described by the variance equation (8), and are independent of N_{rep} .

4.4 Steady-State Validation (Edges 2 & 3)

Edges 2 and 3 of Section 4.1 concern the stationary regime. This section tests both, and checks that all three descriptions reduce correctly to the known limits at the boundaries of parameter space.

The ODE solutions in Figure 4 plateau at the analytical fixed points (14), and the steady-state sampler reproduces the same values to within Monte Carlo error. Figure 7 makes the convergence quantitative: on log-log axes the mean absolute deviation of the sampler’s μ_R^{ss} estimate has a fitted slope of -0.505 ± 0.008 over four decades, against the $-1/2$ of the Central Limit Theorem. Unlike the Edge-1 fit, all 25 points here average over the same number of trials, so the fit needs no reweighting.

A stricter check asks the sampler to reproduce *all five* steady-state moments at once. Table 2 compares the sampled moments against the closed forms (11) and (14) in an asymmetric regime.

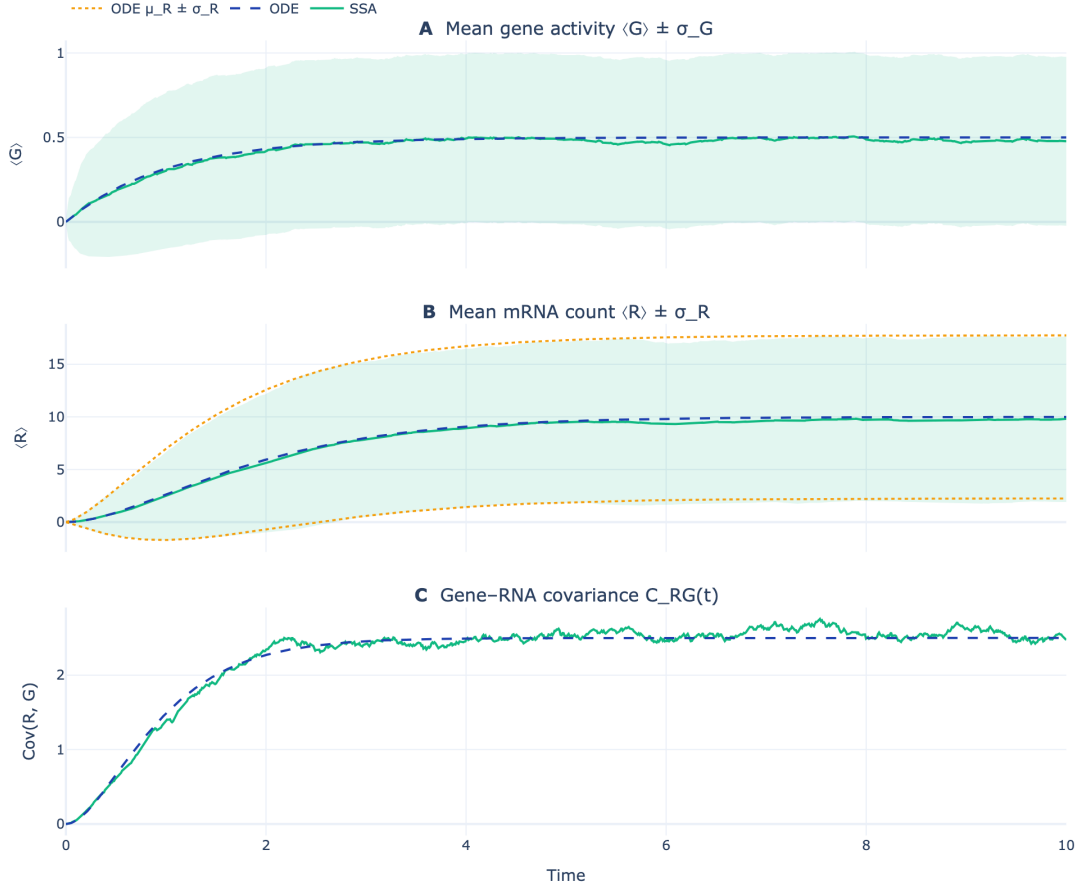


Figure 4: SSA ensemble averages (green, with empirical $\pm\sigma$ fluctuation bands shaded) and the deterministic ODE solutions (dashed blue) at the reference parameters. In the mRNA panel, the analytical spread from the moment ODEs, $\mu_R \pm \sigma_R$, is drawn as amber dotted lines on top of the SSA band.

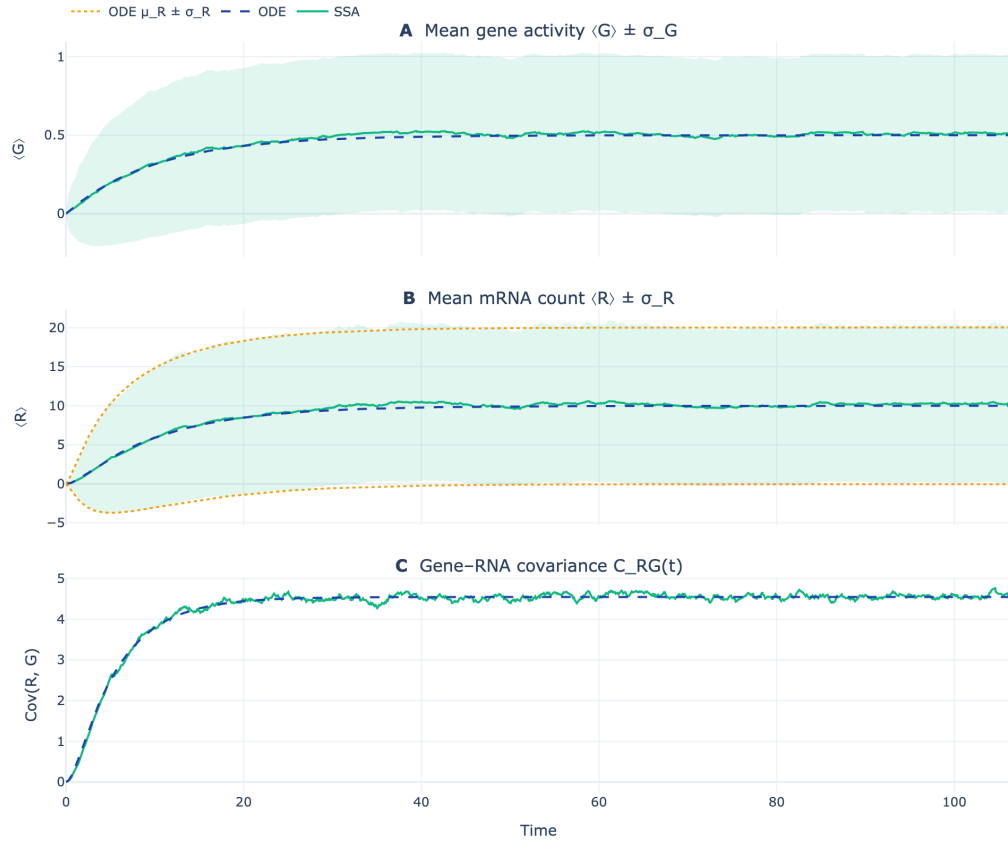


Figure 5: SSA versus moment ODEs in the high-noise slow-switching regime; formatting follows Figure 4. Edge 1 holds beyond the symmetric reference case.

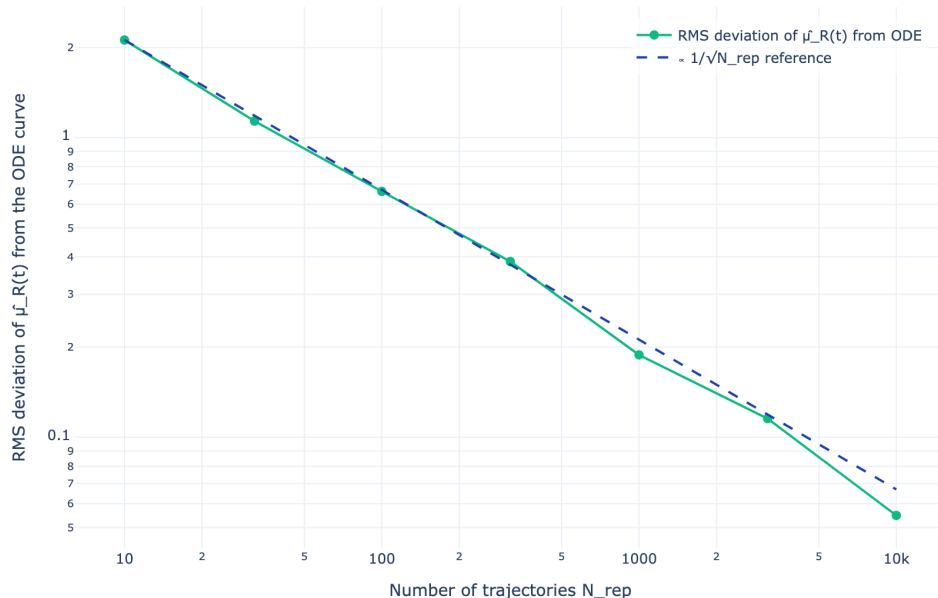


Figure 6: Convergence of the SSA ensemble mean to the moment-ODE solution (Edge 1) at the reference parameters. The statistic is the RMS deviation of $\hat{\mu}_R(t)$ from the ODE curve over the full time grid; the dashed line is a $1/\sqrt{N_{\text{rep}}}$ reference anchored at the smallest ensemble. This is the Edge 1 counterpart of Figure 7, which tests Edge 2.

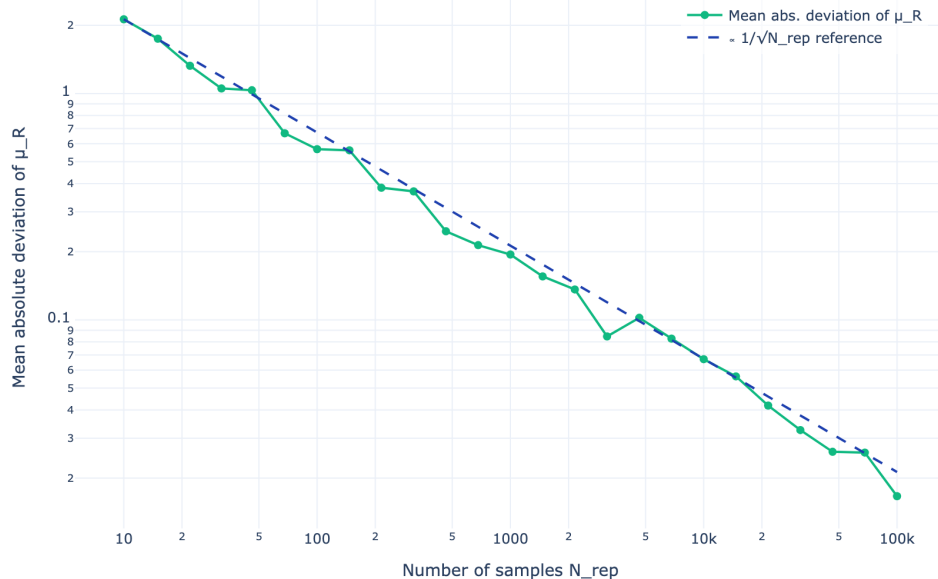


Figure 7: Convergence of the steady-state sampler's empirical moments to the analytical steady-state values as a function of the ensemble size N_{rep} . The error (mean absolute deviation) decays as $1/\sqrt{N_{\text{rep}}}$, the rate given by the Central Limit Theorem.

All five agree to within a few Monte Carlo standard errors.

The five rows are not five independent confirmations. All five moments are computed from a single array of Beta draws, so their deviations are strongly correlated.

Moment	Analytic	Sampler	Abs. diff.	Std. error
μ_G^{ss}	0.3333	0.3317	0.0016	0.0010
μ_R^{ss}	6.6667	6.6277	0.0389	0.0142
σ_G^{ss}	0.4714	0.4708	0.0006	0.0004
σ_R^{ss}	6.4979	6.4767	0.0212	0.0105
C_{RG}^{ss}	1.7778	1.7666	0.0111	0.0071

Table 2: Steady-state moments from the sampler ($N_{\text{rep}} = 2 \times 10^5$) against exact analytical values for $k_{\text{on}} = 0.5, k_{\text{off}} = 1, k_{\text{syn}} = 20, k_{\text{deg}} = 1$. The final column is the Monte Carlo standard error, obtained by bootstrapping over 400 resamples.

The boundary limits predicted in Section 3.3 also hold. At $k_{\text{off}} = 0$ the model reduces to a birth–death process with a Poisson steady state—setting $k_{\text{off}} = 0$ in (13) kills the excess term and leaves $F^{ss} = 1$ exactly: the moment ODEs relax to $\mu_G^{ss} = 1$ and $\mu_R^{ss} = k_{\text{syn}}/k_{\text{deg}}$, and the sampled counts overlay the analytic Poisson PMF (Figure 9) with a measured Fano factor of 0.9953 ± 0.0043 against the theoretical 1. At the opposite boundary, $k_{\text{deg}} = 0$, mRNA accumulates without bound and no steady state exists.

Edge 3 as a distributional test. Since both endpoints of Edge 3 produce samples, the full stationary laws can be compared directly, using the Total Variation distance $\text{TV} = \frac{1}{2} \sum_r |\hat{P}^{\text{SSA}}(r) - \hat{P}^{\text{smp}}(r)|$. On the SSA side the mRNA count is recorded at one late time point per trajectory, which costs samples but avoids the autocorrelation that would understate the spread.

The null distribution for TV is obtained by *direct resimulation*: assuming both methods sample the same stationary law, we draw 200 fresh pairs at the same sample sizes and recompute the statistic. We avoid bootstrapping each side from its own empirical measure, which would add sampling noise twice and centre the null band above the true expected TV, making the test too lenient.

The two regimes in Figure 8 put very different demands on the test. While the symmetric reference regime (left) presents a relatively standard distribution, the slow-switching regime (right) is highly bimodal. A method could match the basic moments μ_R and σ_R exactly and still miss this complex shape, so achieving agreement here rigorously compares the two implementations where the distribution is furthest from anything the moments describe.

Measured by total variation (TV) distance, the two laws strongly agree. In the slow-switching regime, $\text{TV} = 0.0117$ ($p = 0.74$), and in the symmetric regime, $\text{TV} = 0.0093$ ($p = 1.00$). Neither value exceeds what sampling variation alone would produce. Consequently, no statistically significant discrepancy between the two laws is detected in either regime.

These checks show the sampler to be an unbiased, $\sqrt{N}$ -consistent estimator of the steady-state moments, so Edge 2 and Edge 3 hold.

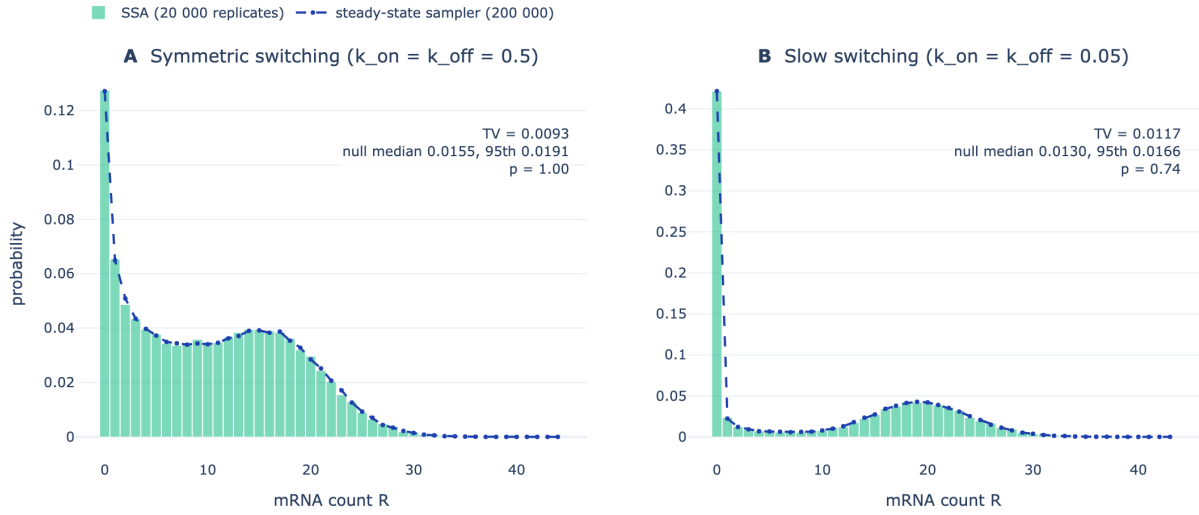


Figure 8: Edge 3 as a distributional test: the stationary mRNA distribution from 20 000 independent SSA trajectories (green bars, one sample per trajectory) against 2×10^5 draws from the steady-state sampler (dashed blue). Left: symmetric reference regime. Right: slow switching ($k_{\text{on}} = k_{\text{off}} = 0.05$), where the distribution is bimodal.

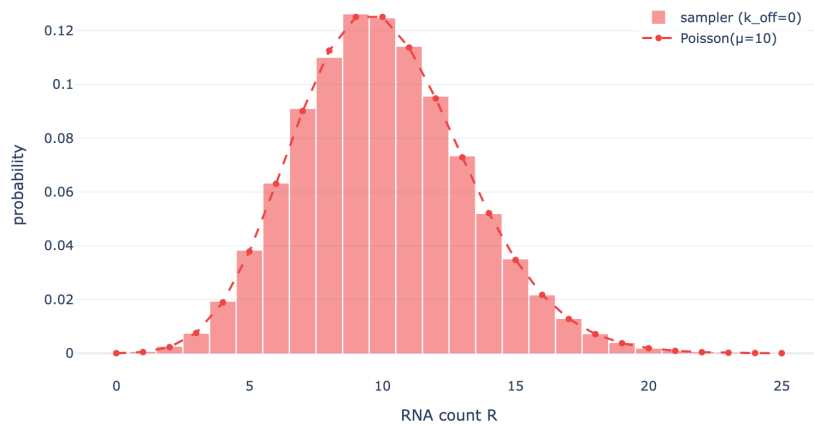


Figure 9: Boundary case $k_{\text{off}} = 0$ (permanently active gene): the sampled steady-state mRNA distribution against the analytic Poisson PMF with mean $k_{\text{syn}}/k_{\text{deg}}$.

5 Conclusion

All three edges of the validation triangle hold. Their value lies in what the three implementations do not share: a trajectory simulator, a deterministic ODE integrator, and a Beta–Poisson sampler have no common code and fail in different ways, so an error in the SSA event loop, a mis-derived moment equation, or a wrong stationary law would each disturb one edge while leaving the others intact. The reductions to classical limits at $k_{\text{off}} = 0$ and $k_{\text{deg}} = 0$ anchor this at the two boundaries where the answer is known in advance.

The triangle validates the three implementations against one another over the regimes tested;

6 Future Work

With the telegraph model established as a validated baseline, two directions follow: one that strengthens the validation, and one that captures more of the underlying biology.

Gene regulatory networks In vivo, gene products often toggle other genes, forming regulatory networks that can be described by mapping the interacting genes to an Ising model, where ON/OFF states act as spins and regulatory links as couplings. The obstacle is the size of the state space: for n genes the joint distribution spans 2^n promoter configurations, each multiplying the unbounded mRNA and protein spaces, so direct integration gives way to approximate high-dimensional density estimators.

Incorporating the protein level. A three-stage model ($\text{gene} \rightarrow \text{mRNA} \rightarrow \text{protein}$) would add translation at rate k_{trans} and protein degradation at rate $k_{\text{deg},p}$. The tools validated here carry over directly: the SSA handles the two additional reactions unchanged, and the moment ODE system expands to cover protein variance and mRNA–protein covariance.

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